CHAPTER 3

THE GREAT DEMOTIONS

> [One philosopher] asserted that he knew the whole secret … [H]e surveyed the two celestial strangers from top to toe, and maintained to their faces that their persons, their worlds, their suns, and their stars, were created solely for the use of man. At this assertion our two travelers let themselves fall against each other, seized with a fit of … inextinguishable laughter.
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> —VOLTAIRE, *MICROMEGAS. A PHILOSOPHICAL HISTORY* (1752)

In the seventeenth century there was still some hope that, even if the Earth was not the center of the Universe, it might be the only "world." But Galileo's telescope revealed that "the Moon certainly does not possess a smooth and polished surface" and that other worlds might look "just like the face of the Earth itself." The Moon and the planets showed unmistakably that they had as much claim to being worlds as the Earth does—with mountains, craters, atmospheres, polar ice caps, clouds, and, in the case of Saturn, a dazzling, unheard-of set of circumferential rings. After millennia of philosophical debate, the issue was settled decisively in favor of "the plurality of worlds." They might be profoundly different from our planet. None of them might be as congenial for life. But the Earth was hardly the only one.

This was the next in the series of Great Demotions, down-lifting experiences, demonstrations of our apparent insignificance, wounds that science has, in its search for Galileo's facts, delivered to human pride.